

Mechanical Systems

Newton's second law to transfer functions: mass-spring-damper systems.

Ogata, *Modern Control Engineering*, Ch. 3.

In this tutorial you will:

- derive the mass-spring-damper model from a free-body diagram,
- see the translational/rotational analogy, and
- build a two-mass system directly in state-space form.

Step through with **Ctrl+Enter**, or render a report with **publish**.

Mechanical systems obey Newton's second law, $\sum F = m\ddot{x}$. For a translational mass-spring-damper system with displacement $x(t)$ driven by force $u(t)$, the free-body diagram gives three opposing forces on the mass: the applied force u , a spring force $-kx$ (Hooke's law), and a damping force $-b\dot{x}$ (viscous friction, proportional to velocity). Summing forces:

$$m\ddot{x} = u - b\dot{x} - kx \quad \Rightarrow \quad m\ddot{x} + b\dot{x} + kx = u$$

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Transfer Function of the Translational System

Taking the Laplace transform with zero initial conditions:

$$(ms^2 + bs + k)X(s) = U(s) \quad \Rightarrow \quad G(s) = \frac{X(s)}{U(s)} = \frac{1}{ms^2 + bs + k}$$

With $m = 1$ kg, $b = 4$ N · s/m, $k = 20$ N/m:

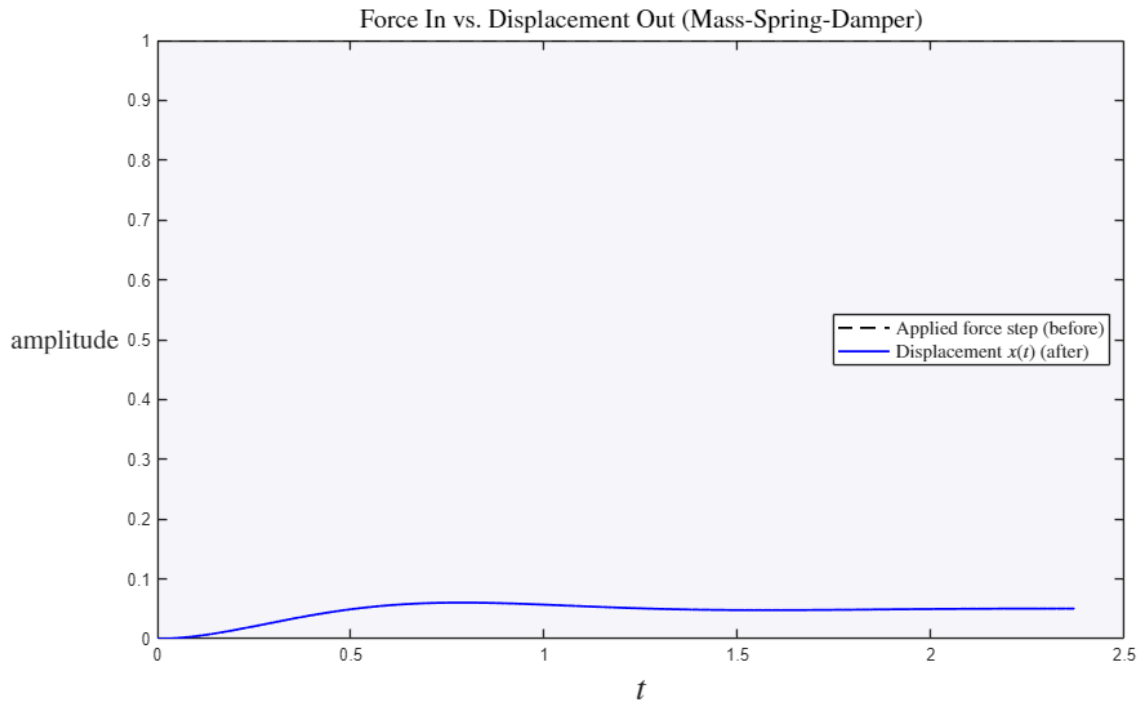
```
m = 1; b = 4; k = 20;
G_trans = tf(1,[m b k])

figure
[xs,ts] = step(G_trans);
plot(ts, ones(size(ts)), 'k--', ts, xs, 'b', 'LineWidth', 1.3)
legend('Applied force step (before)', 'Displacement $x(t)$ (after)', ...
       'Interpreter', 'latex', 'FontSize', 11, 'Location', 'east')
title('Force In vs. Displacement Out (Mass-Spring-Damper)', 'Interpreter', 'latex', 'FontSize', 15)
ylabel('amplitude', 'Interpreter', 'latex', 'FontSize', 16)
set(get(gca, 'YLabel'), 'Rotation', 0)
xlabel('$t$', 'Interpreter', 'latex', 'FontSize', 20)
```

```
G_trans =
```

```
      1
-----
s^2 + 4 s + 20
```

Continuous-time transfer function.



Rotational Analog

The rotational analog replaces force with torque T , mass with moment of inertia J , viscous friction coefficient b , and linear spring constant k with torsional spring constant k . Newton's second law for rotation, $\sum T = J\ddot{\theta}$, gives

$$J\ddot{\theta} + b\dot{\theta} + k\theta = T$$

which is structurally identical to the translational equation under the analogy $x \leftrightarrow \theta$, $m \leftrightarrow J$, $u \leftrightarrow T$.

```
J = 0.5; b_rot = 1; k_rot = 4;
G_rot = tf(1,[J b_rot k_rot])
```

```
G_rot =
```

$$\frac{1}{0.5 s^2 + s + 4}$$

```
Continuous-time transfer function.
```

Two-Mass System

A more complex example: two masses m_1, m_2 connected by a spring k and damper b between them, with force u applied to m_1 and m_2 free to move. Free-body diagrams for each mass give

$$m_1\ddot{x}_1 = u - k(x_1 - x_2) - b(\dot{x}_1 - \dot{x}_2)$$

$$m_2\ddot{x}_2 = k(x_1 - x_2) + b(\dot{x}_1 - \dot{x}_2)$$

In state-space form with state $z = [x_1 \ \dot{x}_1 \ x_2 \ \dot{x}_2]^T$:

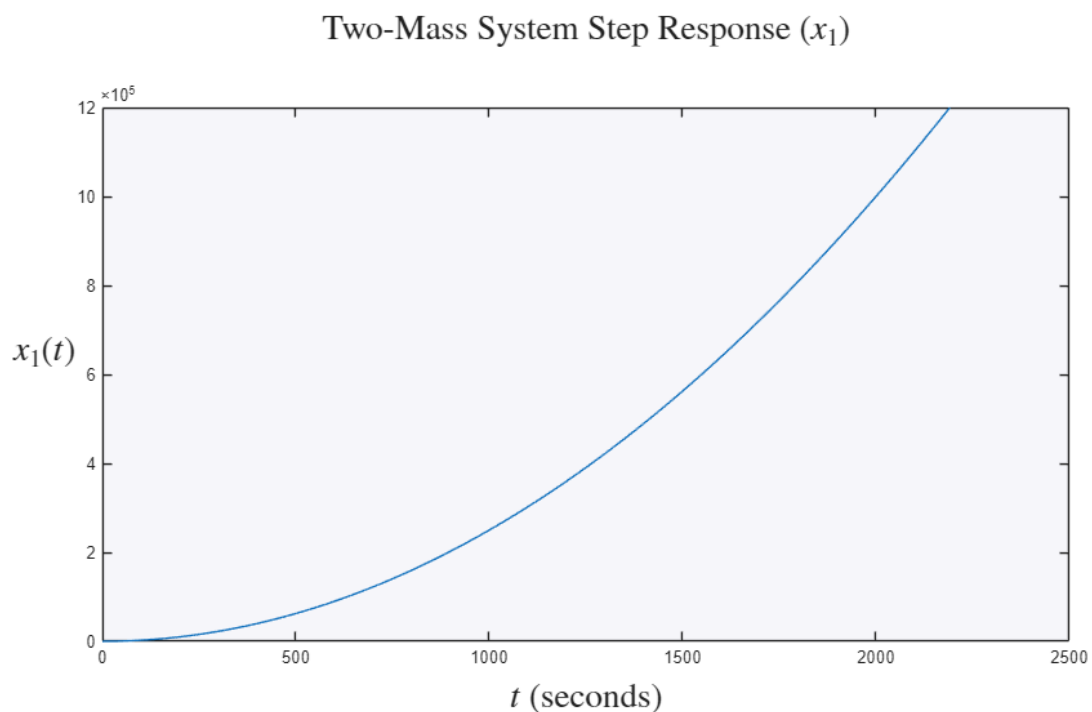
```
m1 = 1; m2 = 1; k2 = 10; b2 = 2;
A = [0 1 0 0
     -k2/m1 -b2/m1 k2/m1 b2/m1
```

```

    0 0 0 1
    k2/m2 b2/m2 -k2/m2 -b2/m2];
B = [0; 1/m1; 0; 0];
C = [1 0 0 0];
D = 0;
sys_2mass = ss(A,B,C,D);

figure
step(sys_2mass)
title('Two-Mass System Step Response ( $x_1$ )','Interpreter','latex','FontSize',20)
ylabel('x_1(t)','Interpreter','latex','FontSize',20)
set(get(gca, 'YLabel'), 'Rotation', 0)
xlabel('t','Interpreter','latex','FontSize',20)

```



Try it yourself

- Lower the damping to $b = 1$ and notice the step response start to overshoot – the same underdamped behavior as a lightly damped circuit.
- Set $k = 0$ (no spring) and watch the system lose its restoring force.